

FDTScore Data Study

PE Factory & Portofio Strategy Analysis

Sainan, Lambert, Xu, Emily, Caleb

Science Department

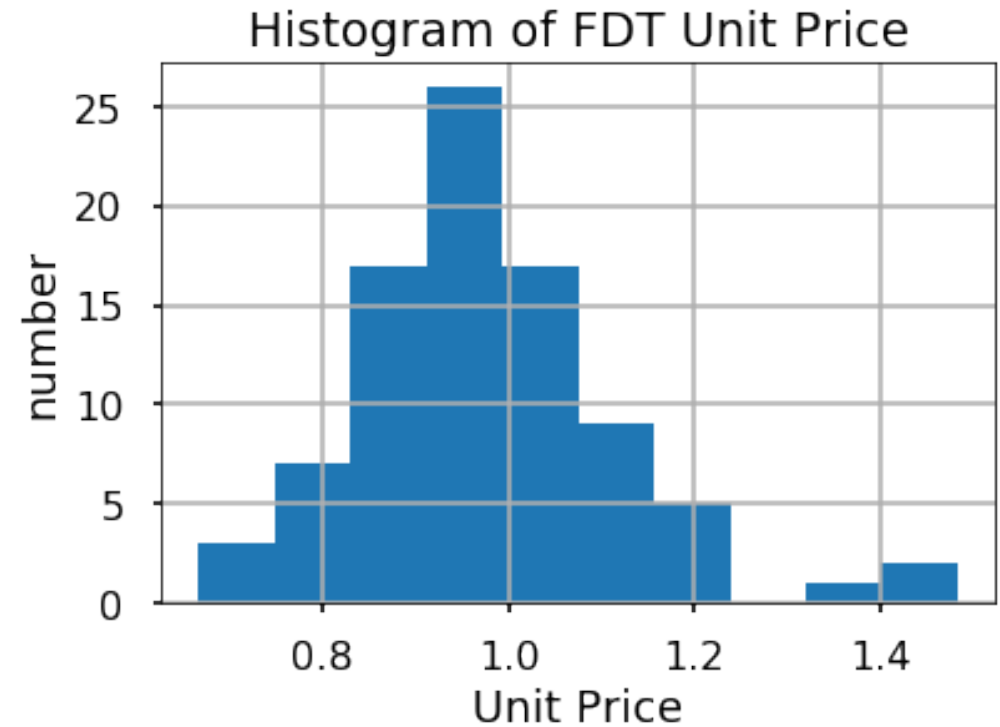
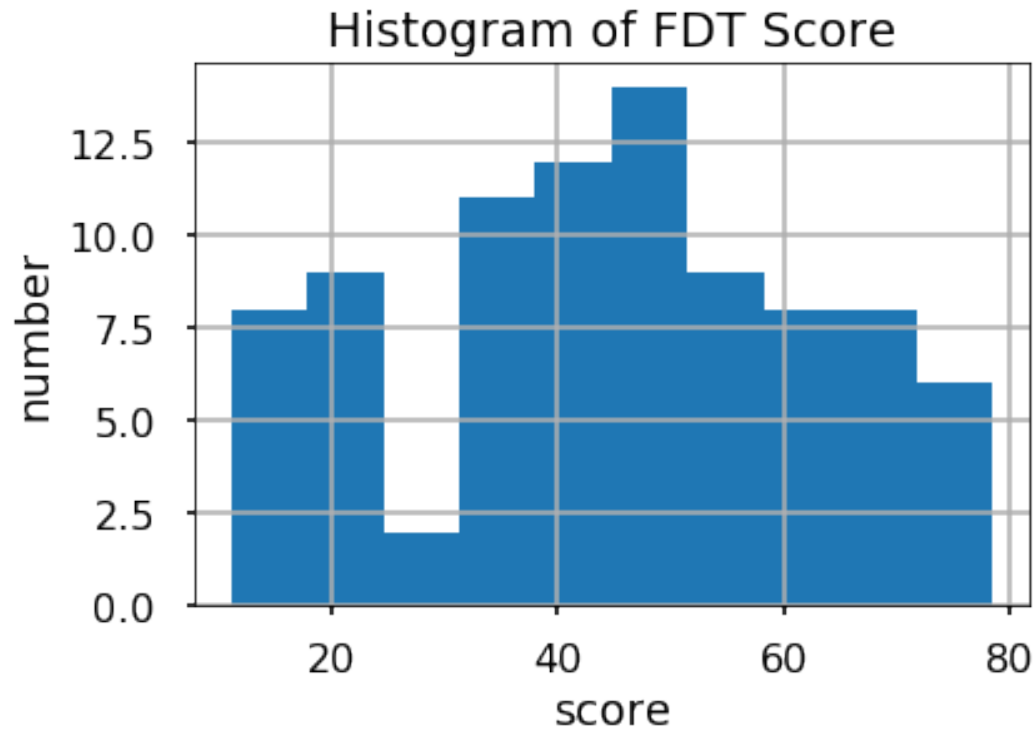
2017-Nov-23

Outline



- **PE Factory with FDTScore**
- Portfolio Strategies Details

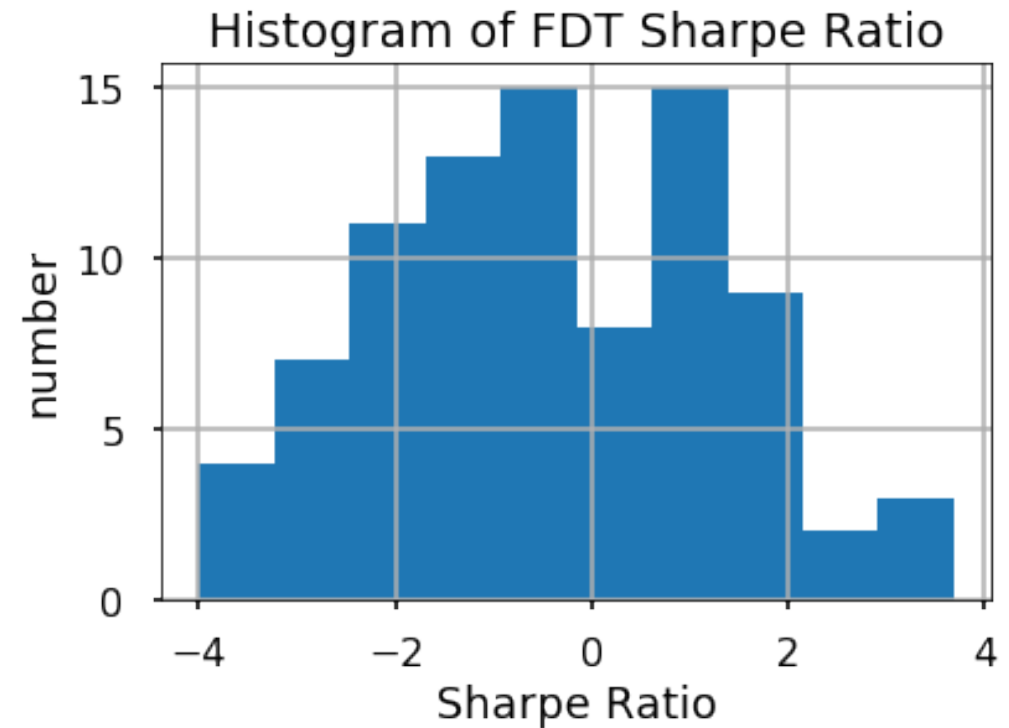
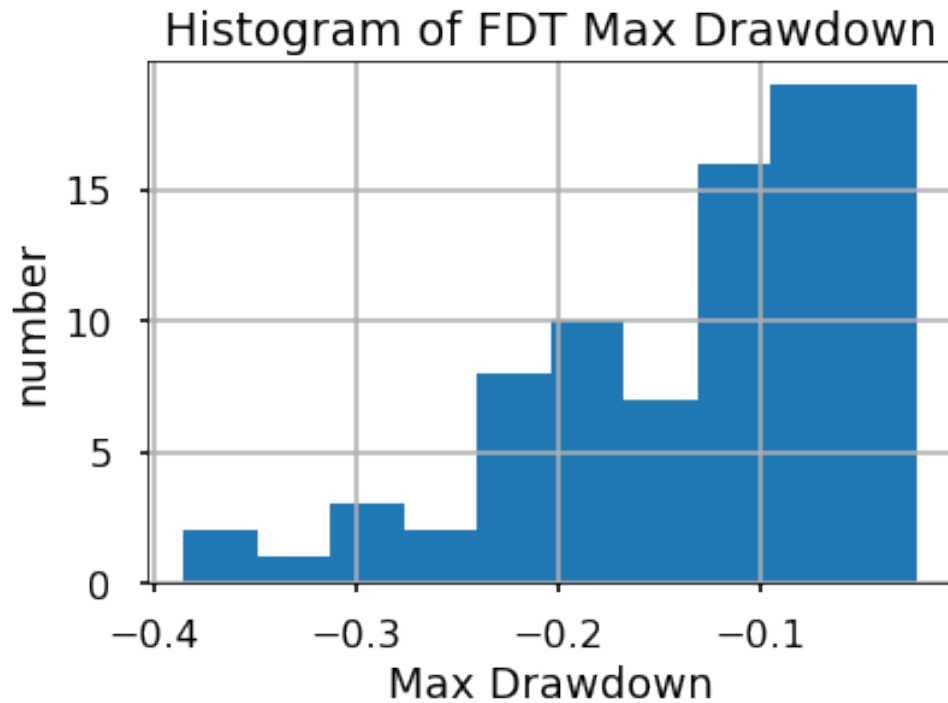
FDTScore Statistics



Overall PnL of all traders up to 2017-11-21

- USD: -107,127.00

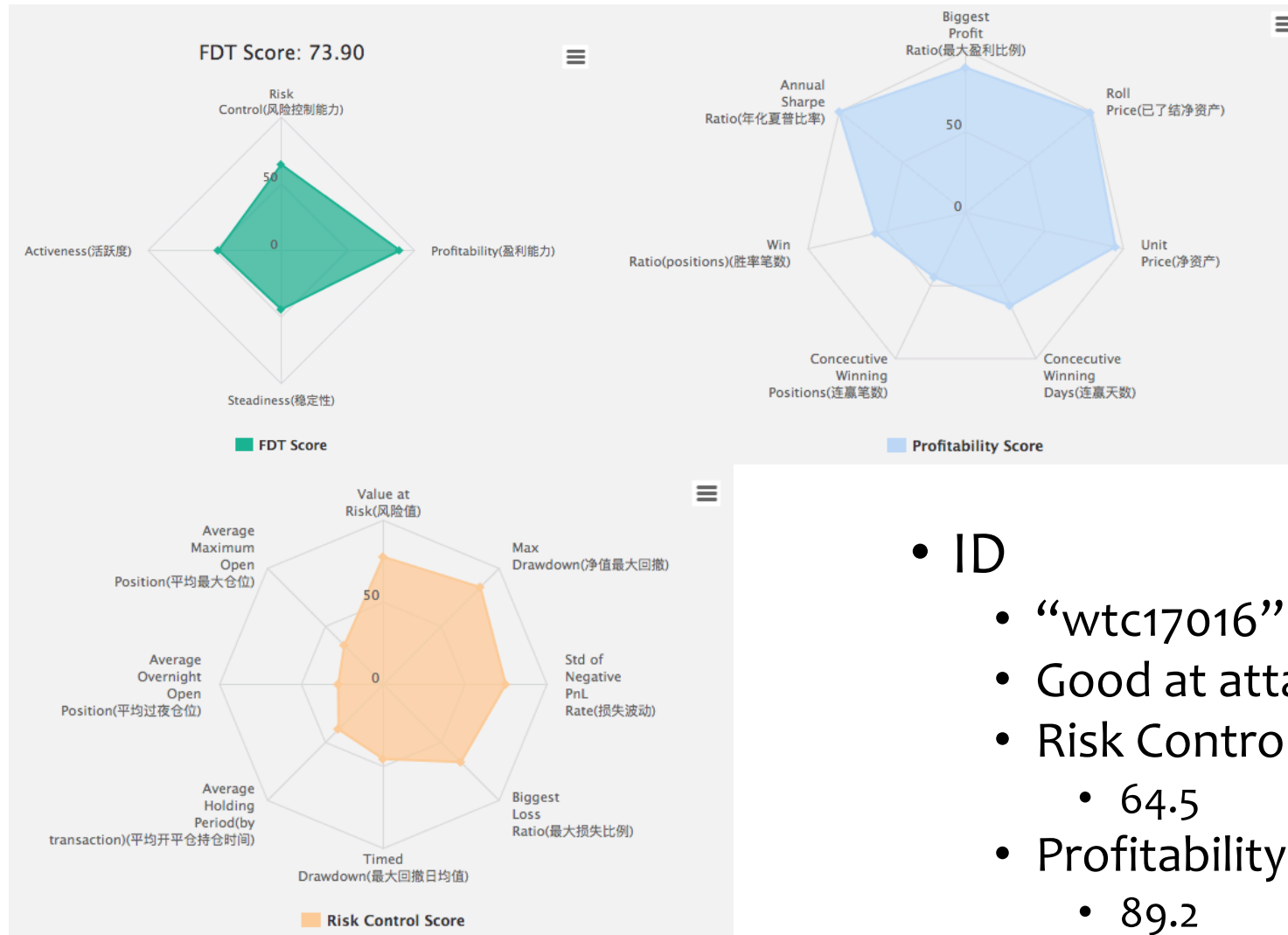
FDTScore Statistics



Talents thru FDTScore



Talents thru FDTScore



- ID
 - “wtc17016”
 - Good at attacking
 - Risk Control
 - 64.5
 - Profitability
 - 89.2

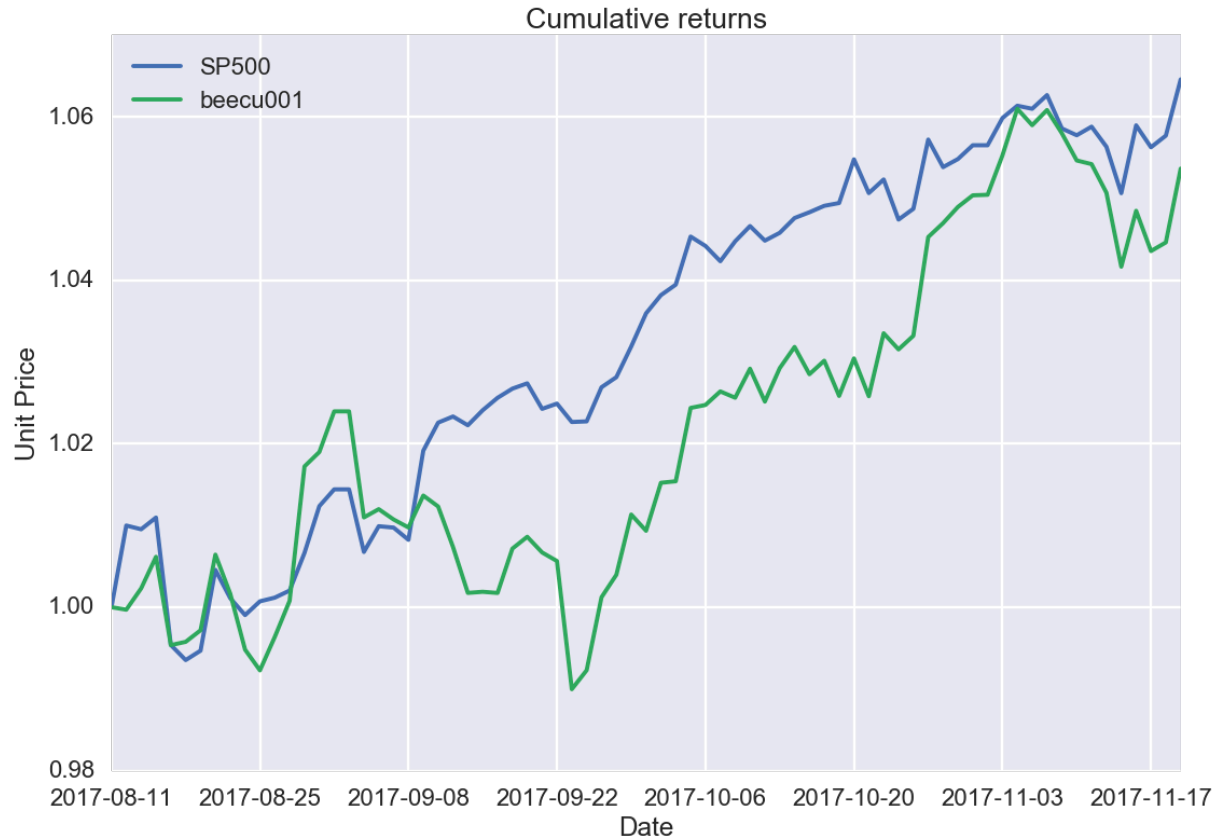
Outline

- PE Factory with FDTScore
- **Detail about Portfolio Strategies**
 - **FDTScore**
 - **Performance Analysis**
 - **Drawdown Analysis**
 - **Diversity Analysis**

Portfolio Strategy “beecu001”

U.S. Stock

FDTAI



• Portfolio “beecu001”

- Started on 2017-08-11 (68 days)
- Market: US Stock
- Benchmark: S&P 500

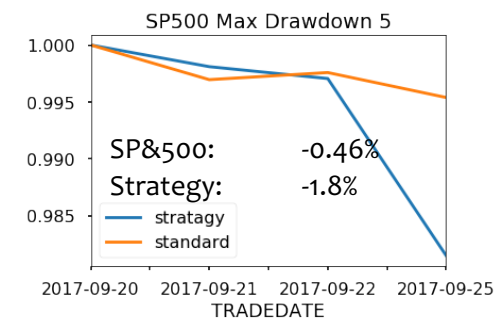
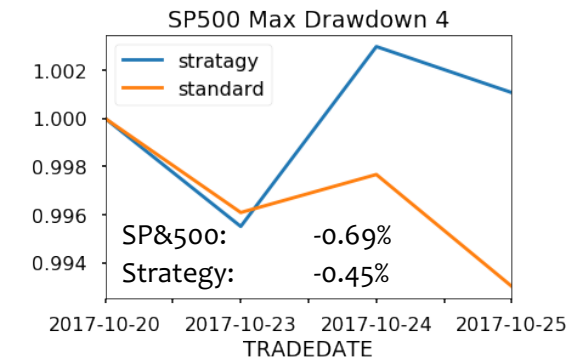
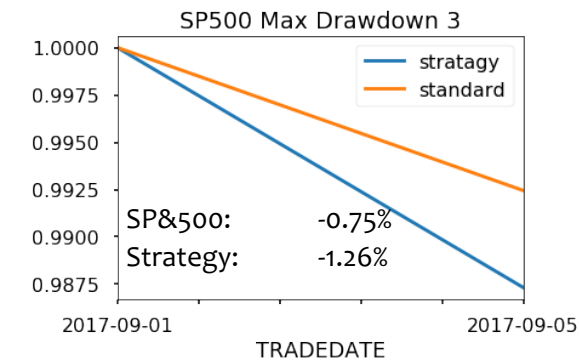
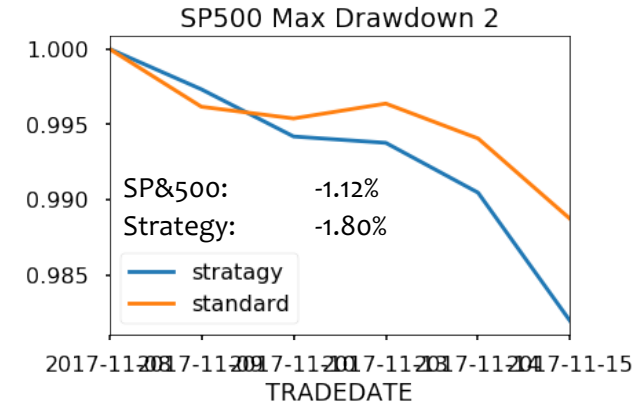
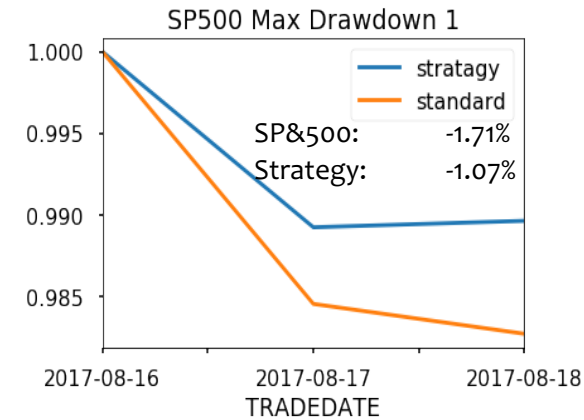
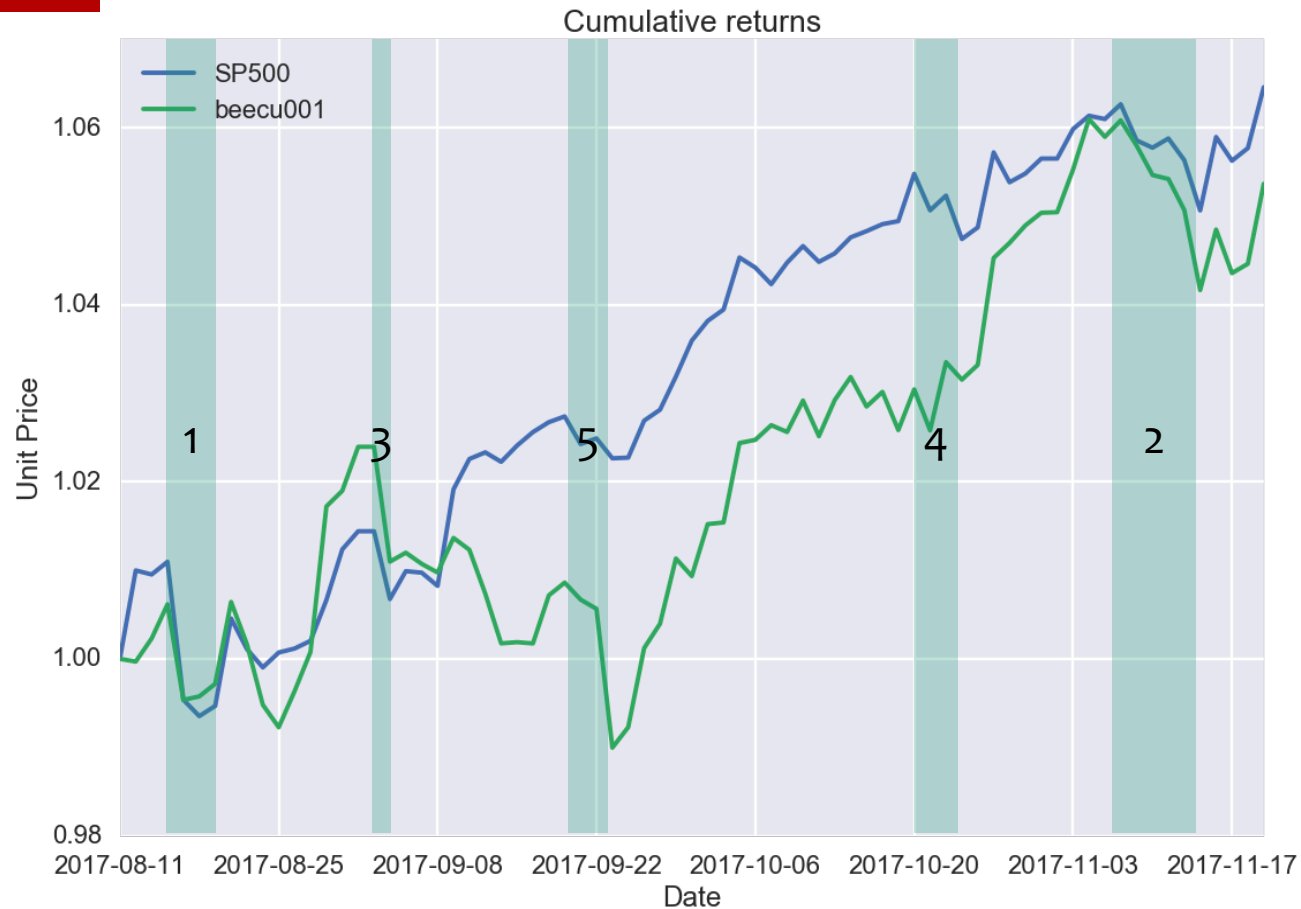
- Alpha: -0.02
- Beta: 1.05

	beecu001	S&P 500
Annual Return	21%	23.7%
Annual Volatility	8.9%	6.3%
Annual Sharpe Ratio	2.2	3.43
Max Drawdown	-3.3%	-1.7%

Portfolio Strategy “beecu001”

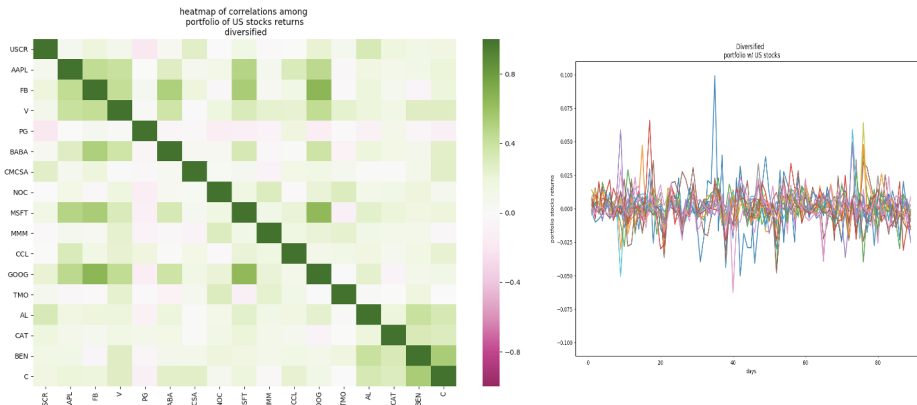
Blue: Strategy
Orange: Benchmark

FDTAI



Diversity: beecu001 U.S. stocks

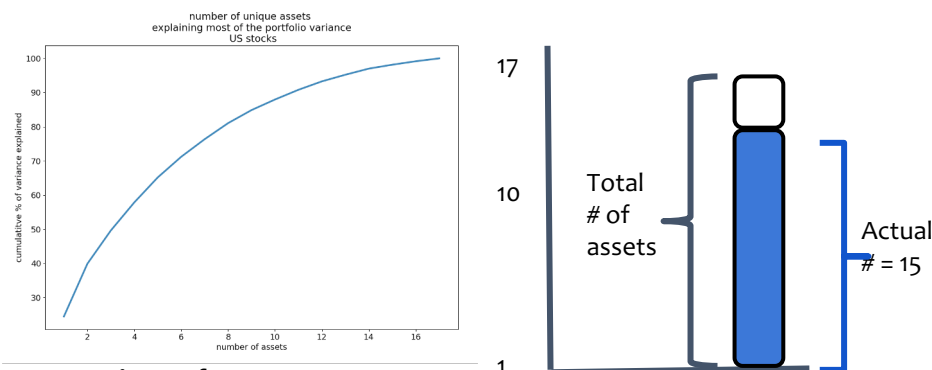
Fig. 1, 2: Portfolio asset similarity



X & Y axes: assets. Asset Correlations: Green = similar; Light = different; Pink = hedge

X-axis: days. Y-axis: asset returns $\in [-10\%, +10\%]$

Fig. 3, 4: Actual # of assets in portfolio



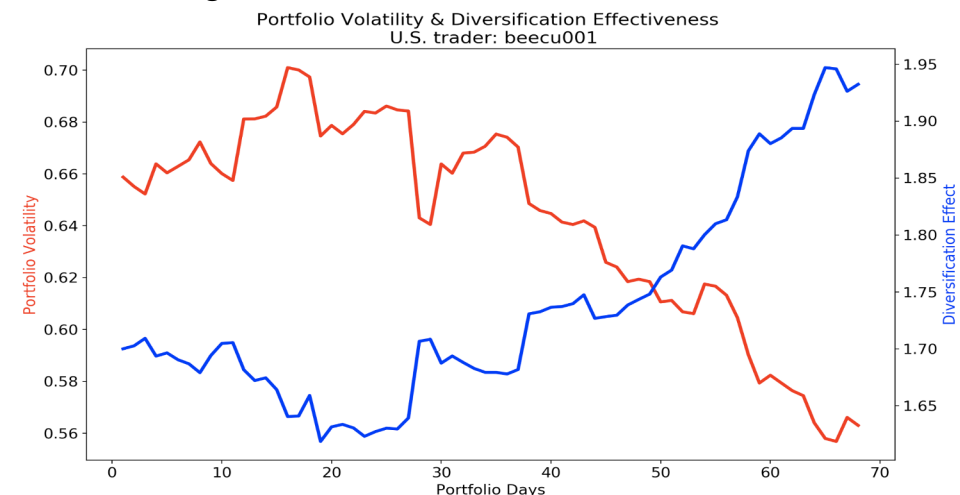
X-axis: # of assets. Y-axis: cumulative % of portfolio variance explained

The goal is to REDUCE RISK from diversification. Thus, the effect of diversification can be measured by this ratio - portfolio's weighted average asset volatility to its actual volatility. This measures the **essence of diversification, because we can quantify the amount of risk reduced by portfolio diversification!**

Since different asset classes are not perfectly correlated to each other, this ratio in general > 1 . In other words, the overall risk of a well-diversified portfolio is less than the weighted-average risk of its component holdings. $DE \in [1, \infty)$, where 1 indicates no diversification effect, larger number means better diversified. $DE = \{ \sum_j^S (W_j * \sigma_j) \} / \sigma_p$

Where σ_p = total portfolio risk = $\sqrt{W^T \text{Cov} W}$, Cov is the variance covariance matrix on daily asset returns, W is vector of weights
 σ_j = asset j risk = $\sqrt{\sum_i^n (j_i - \mu_j) / n}$, j_i is return of asset j on day i

Fig. 5 Portfolio Diversity Risk Reduction Effect



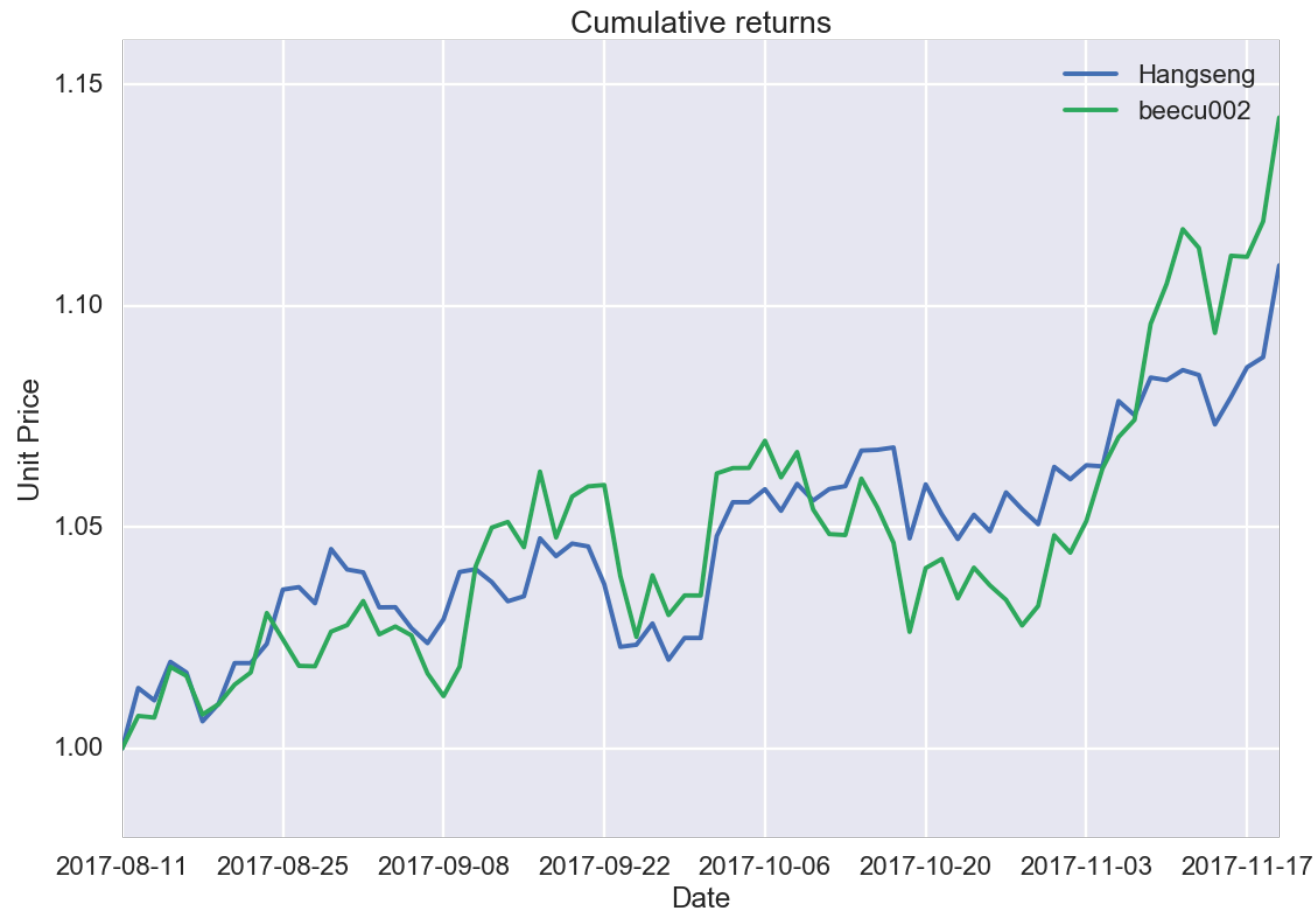
X-axis: days. Red Y-axis: portfolio risk. Blue Y-axis: DE

Overall risk was reduced by 195%!

Portfolio risk: 0.5 - 0.7 DE: 1.6-1.95

Portfolio Strategy “beecu002”

HK Stock



• Portfolio “beecu002”

- Started on 2017-08-11 (68 days)
- Market: HK Stock
- Benchmark: HS Index

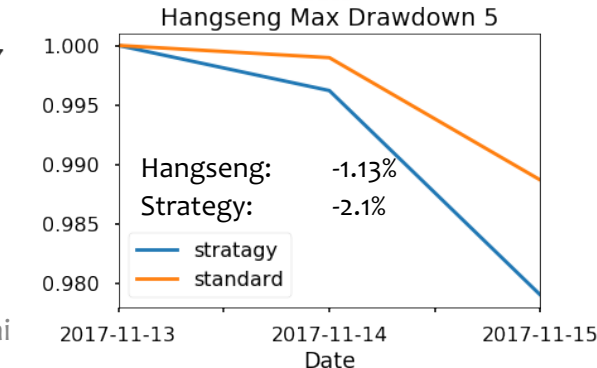
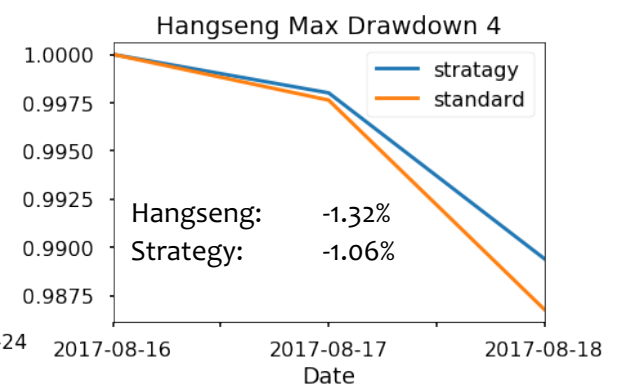
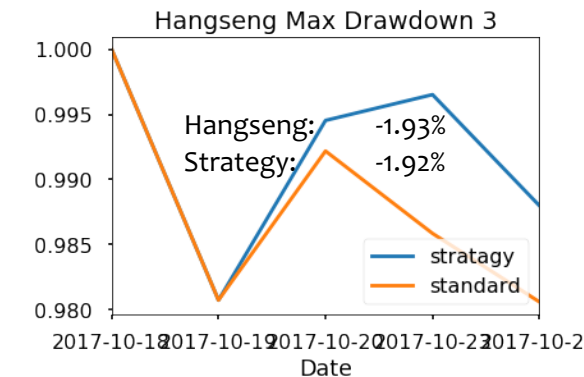
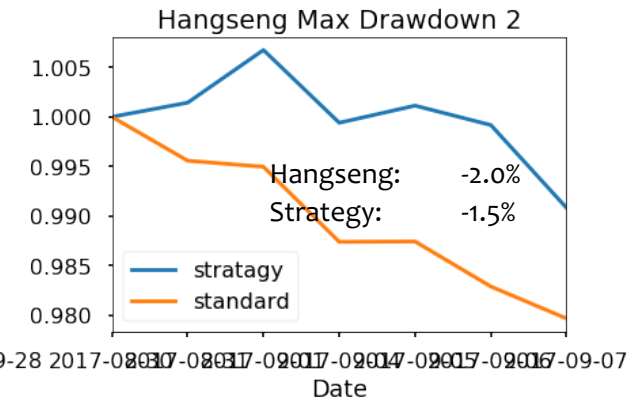
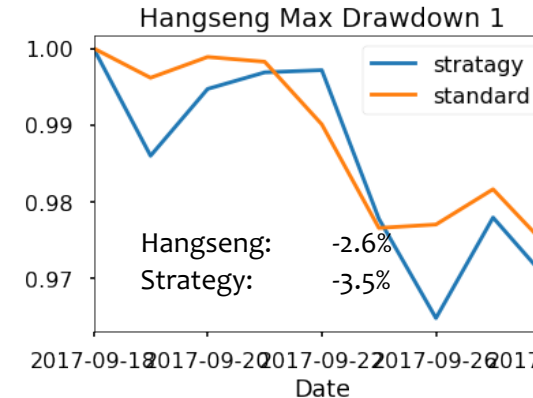
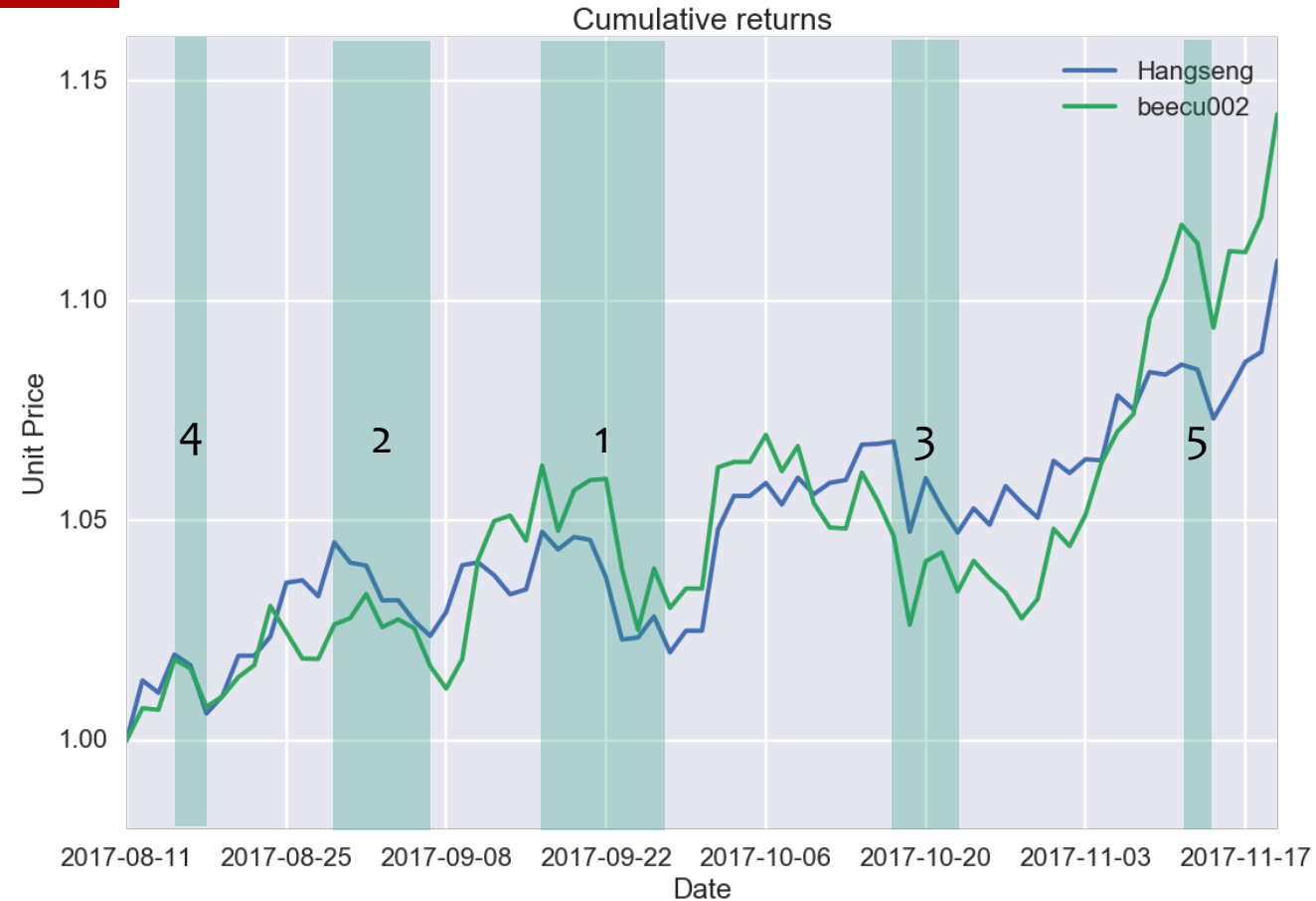
- Alpha: 0.15
- Beta: 0.95

	beecu002	HS Index
Annual Return	58.4%	38.9%
Annual Volatility	15.7%	11.7%
Annual Sharpe Ratio	3.01	2.87
Max Drawdown	-4%	-2.6%

Portfolio Strategy “beecu002”

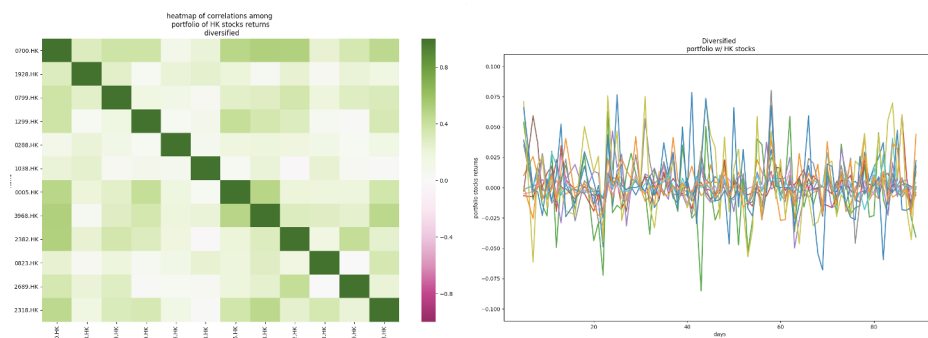
Blue: Strategy
Orange: Benchmark

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Diversity: beecu002 HK stocks

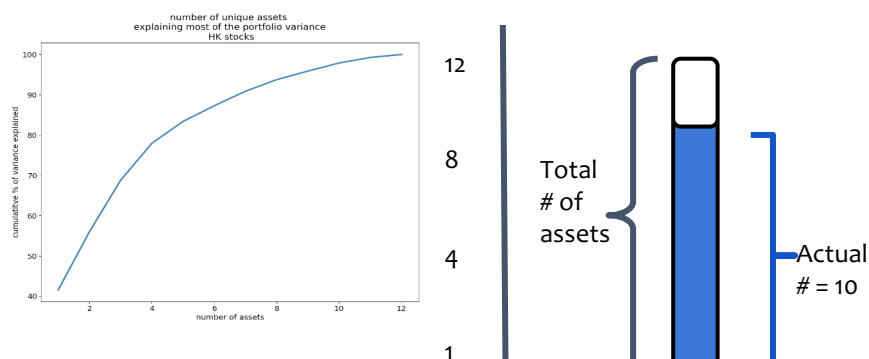
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X & Y axes: assets. Asset Correlations: green = similar; Light = different; Pink = hedge

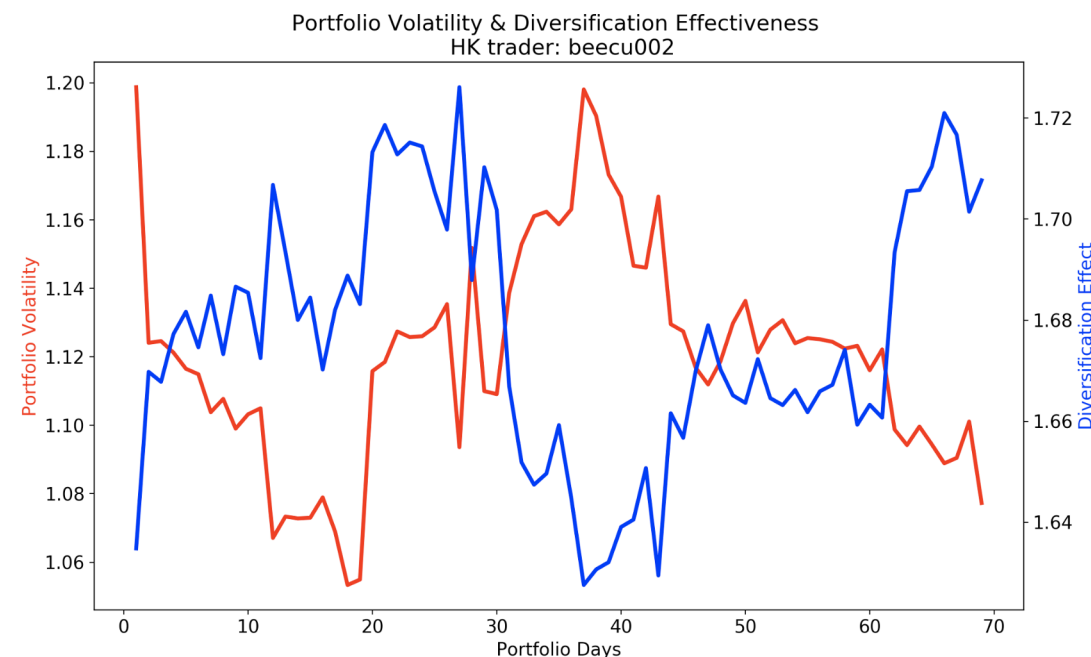
X-axis: days. Y-axis: asset returns $\in [-10\%, +10\%]$

Fig. 3, 4: Actual # of assets in portfolio



X-axis: # of assets. Y-axis: cumulative % of portfolio variance explained

Fig. 5 Portfolio Diversity Risk Reduction Effect



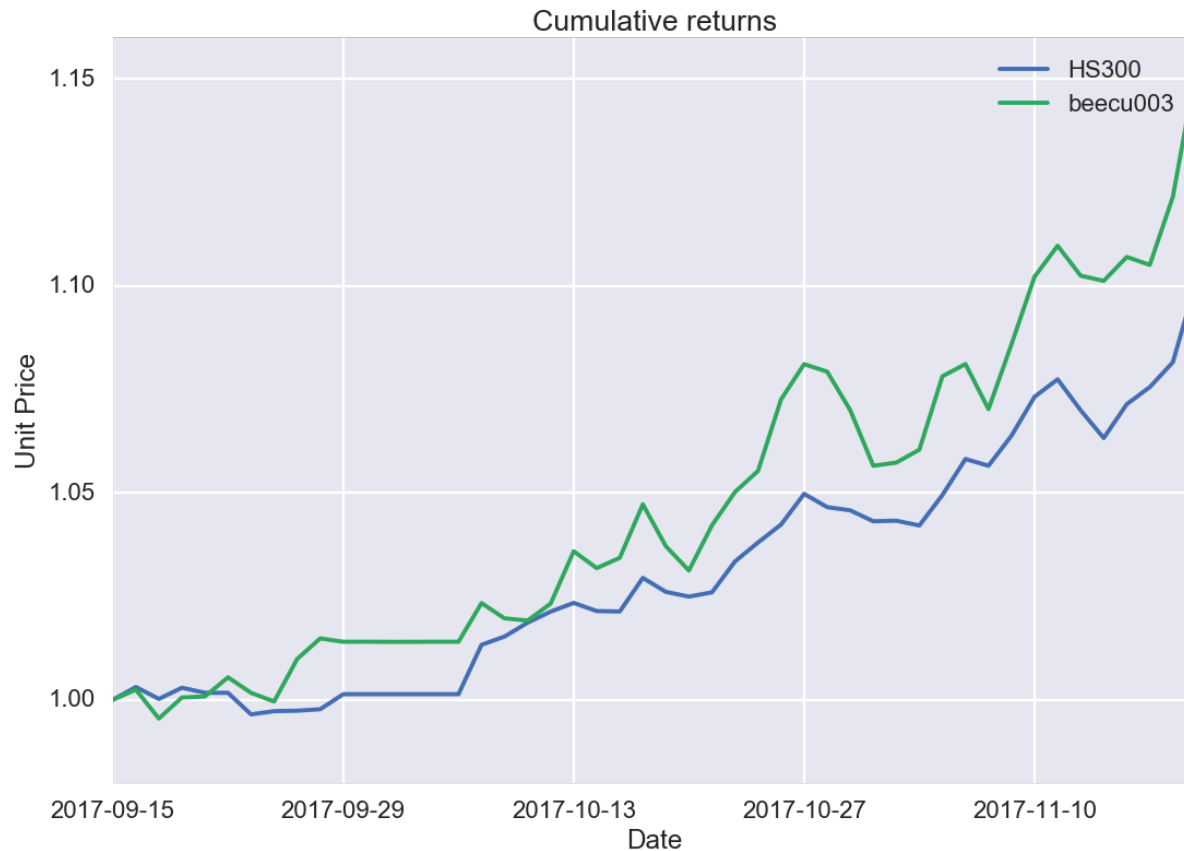
X-axis: days. Red Y-axis: portfolio risk. Blue Y-axis: DE

Overall risk was reduced by 170%

Portfolio risk: 1 - 1.2 DE: 1.6 - 1.7

Portfolio Strategy “beecu003”

China A-share



• Portfolio “beecu003”

- Started on 2017-09-15 (43 days)
- Market: China-A
- Benchmark: HS300

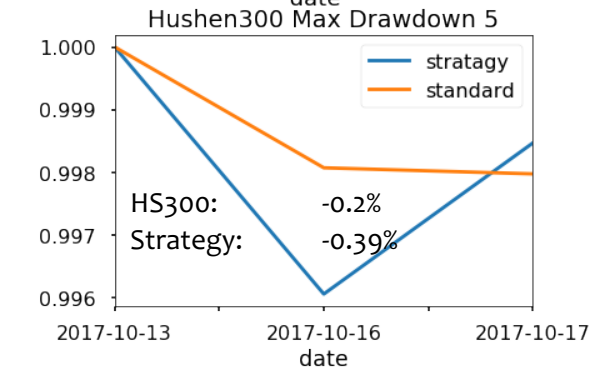
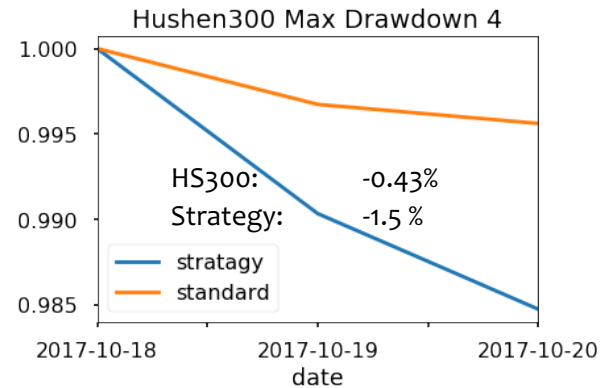
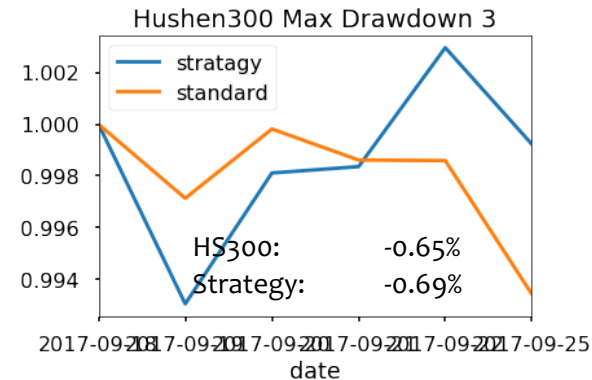
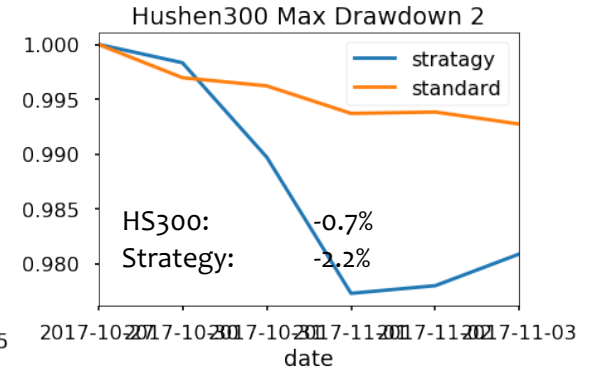
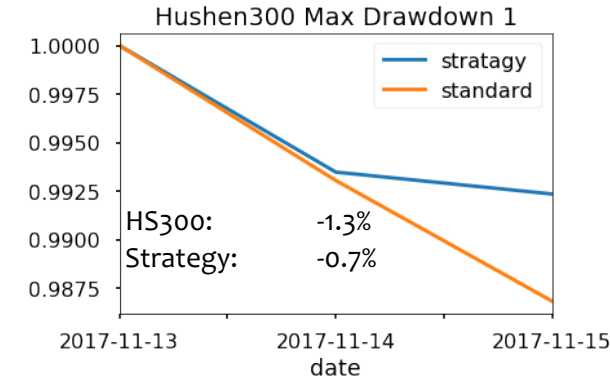
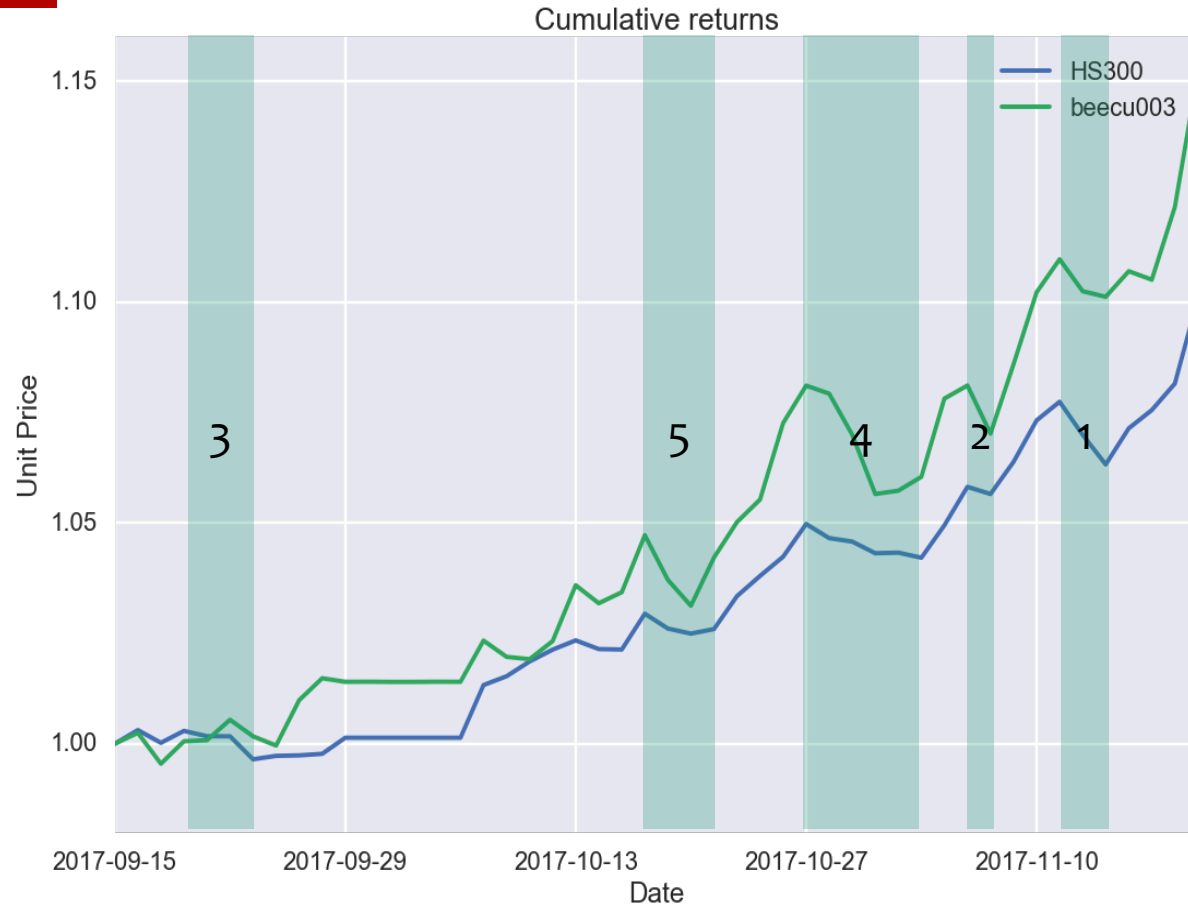
- Alpha: 0.10
- Beta: 1.30

	beecu003	沪深300
Annual Return	33.7%	67.2%
Annual Volatility	13.6%	8%
Annual Sharpe Ratio	5	6.5
Max Drawdown	-2.3%	-1.3%

Portfolio Strategy “beecu003”

Blue: Strategy
Orange: Benchmark

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Diversity: beecu003 A stocks

Fig. 1, 2: Portfolio asset similarity

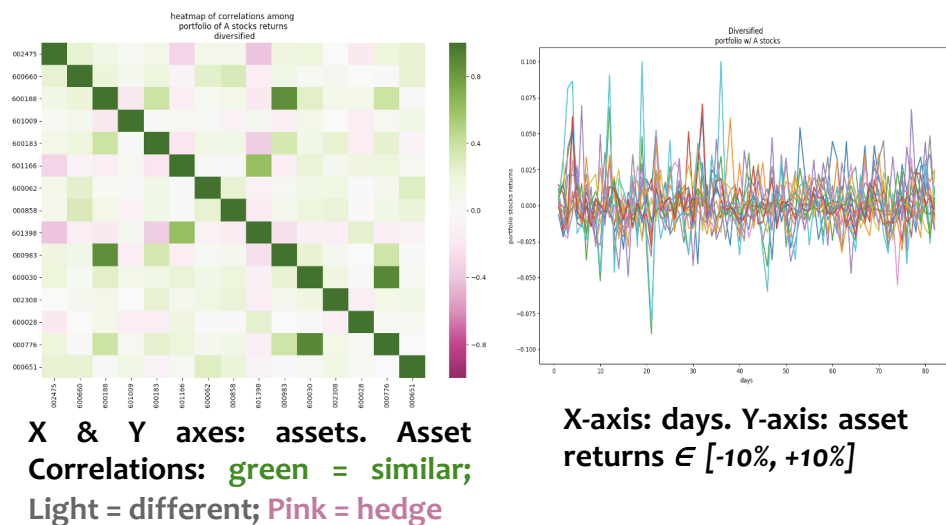


Fig. 3, 4: Actual # of assets in portfolio

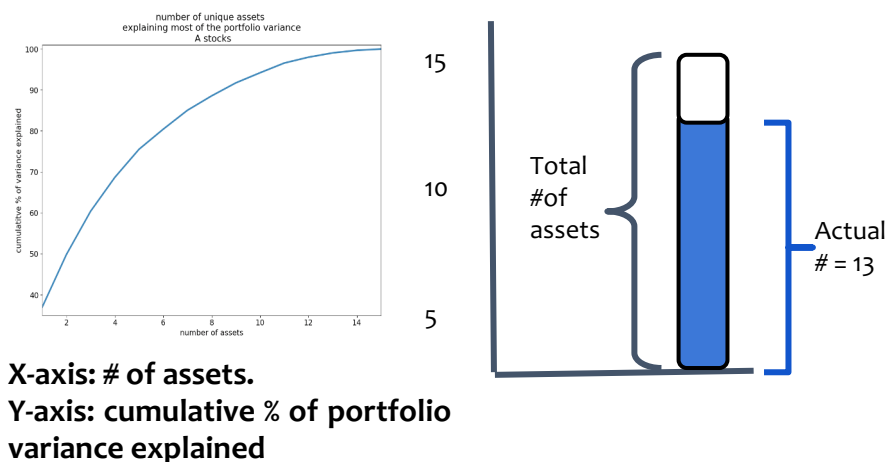
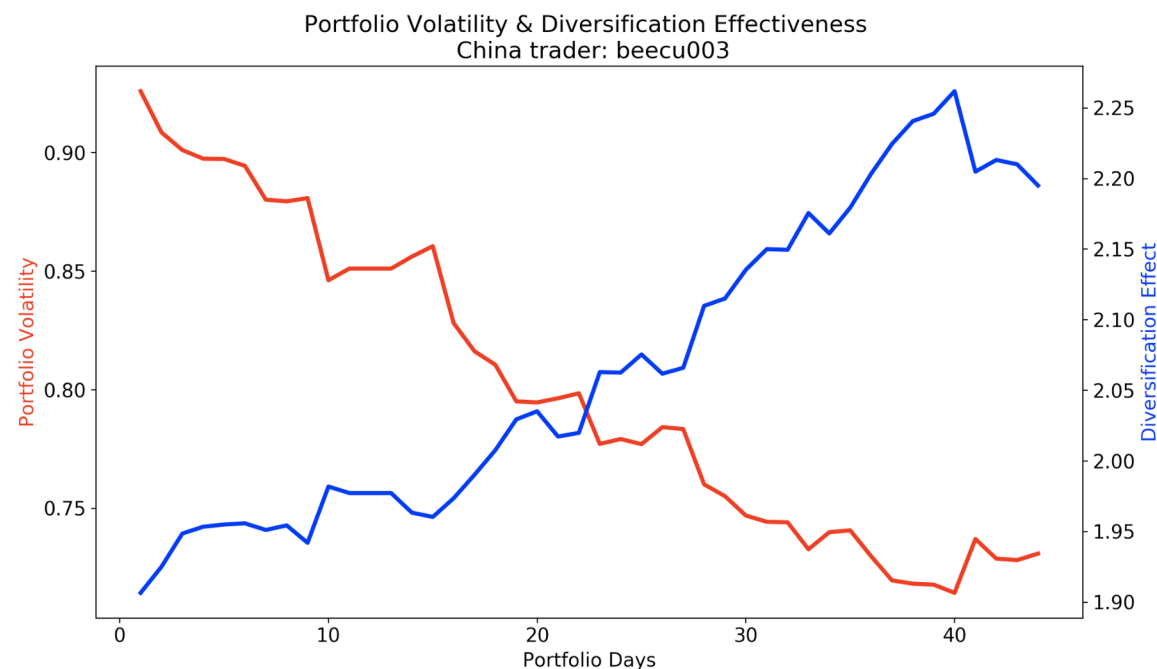


Fig. 5 Portfolio Diversity Risk Reduction Effect



X-axis: days. Red Y-axis: portfolio risk. Blue Y-axis: DE

Overall risk was reduced by 225%!

Portfolio risk: 0.7 - 0.95 DE: 1.9 - 2.25

Diversity Analysis Summary

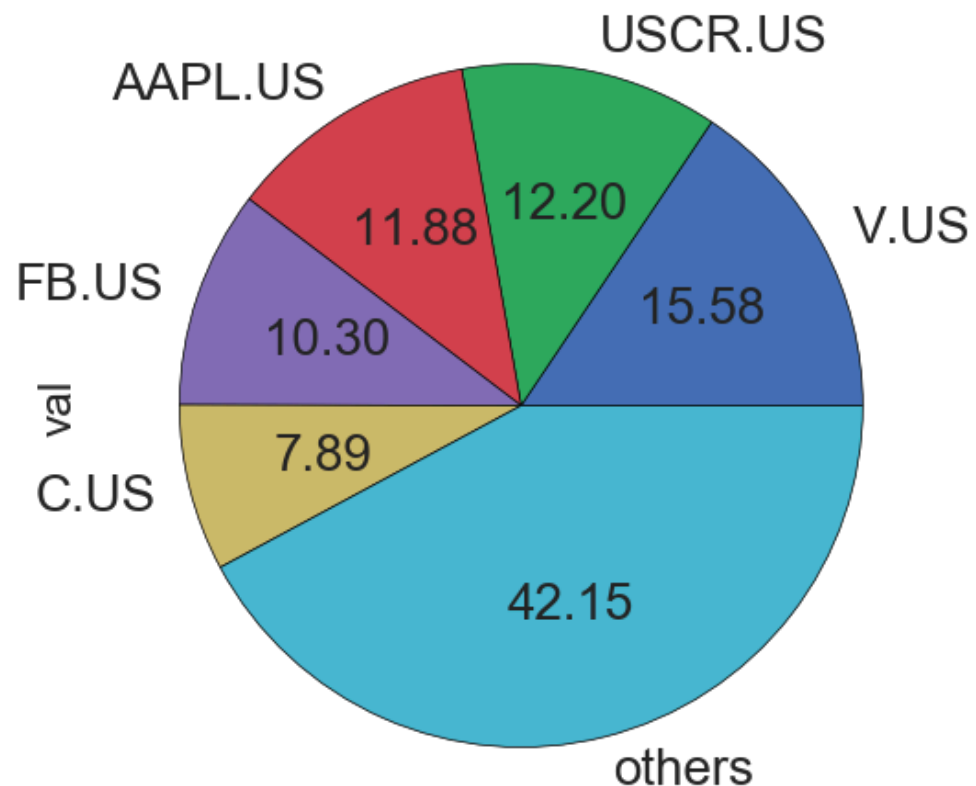
Asset similarity

- The trader in **A stocks** ('beecu003') market is the most-diversified, even showing strategies of hedging.
- The trader in **US stocks** ('beecu001') market is well-diversified, but not as good in A stocks market; there is less hedging.
- The trader in **HK stocks** ('beecu002') market is the least-diversified, compared to the other two. As shown by the heatmap, similarities are higher among the assets.

Diversity Effect in Risk Reduction

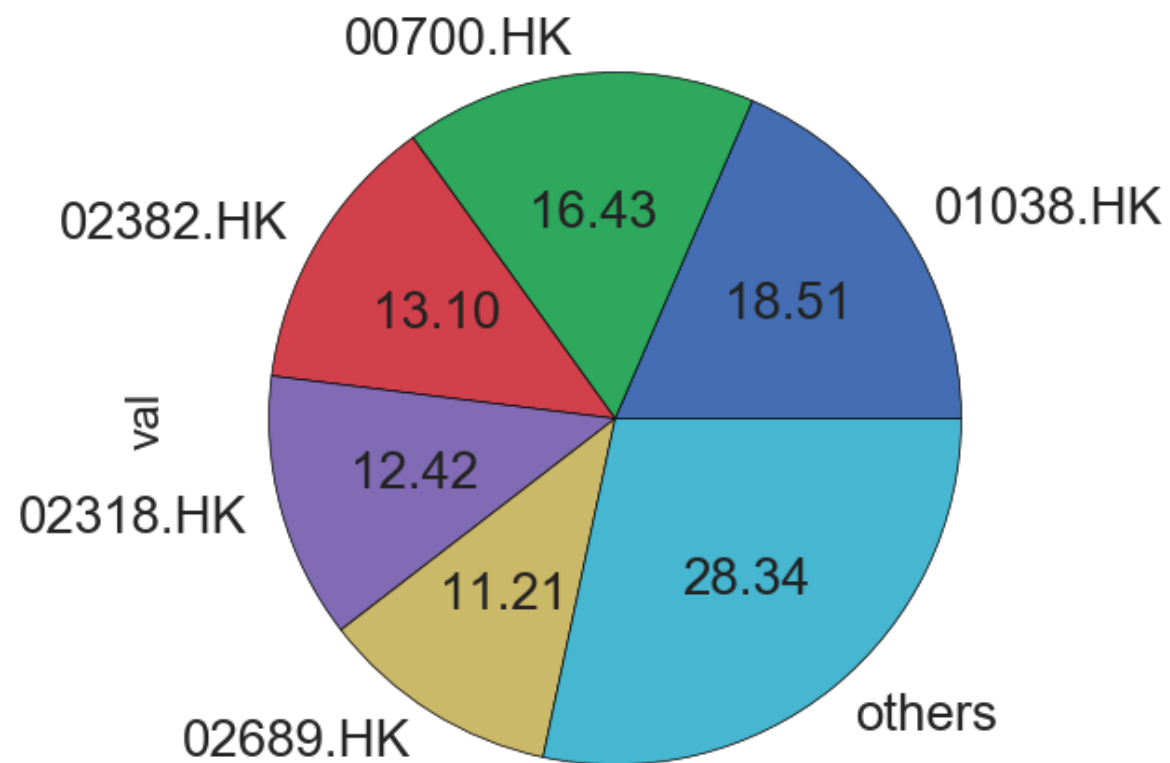
- beecu003 (A stocks): best trend - gradual and sustained improvement in DE. Portfolio is well adjusted for increasing diversification, and risk decreases over time.
- beecu001 (US stocks) also show this trend. **Their portfolio asset choices and weightings are chosen correctly over time.**
- beecu002 (HK stocks), DE more random and risk is volatile throughout. **Trader should to select his asset choices and weights better.**
- beecu003 in **A stocks** reduced risk *the most*, by 225%, and risk ranged from 0.70-0.95.
- beecu001 in **US stocks** reduced risk by 195%, and risk ranged from 0.56-0.70.
- beecu002 in **HK stocks** reduced risk *the least*, by 170%, and risk ranged from 1-1.2.
- This overall risk is lowest in US market and highest in HK market.

Heavy-position Assets – US Stock



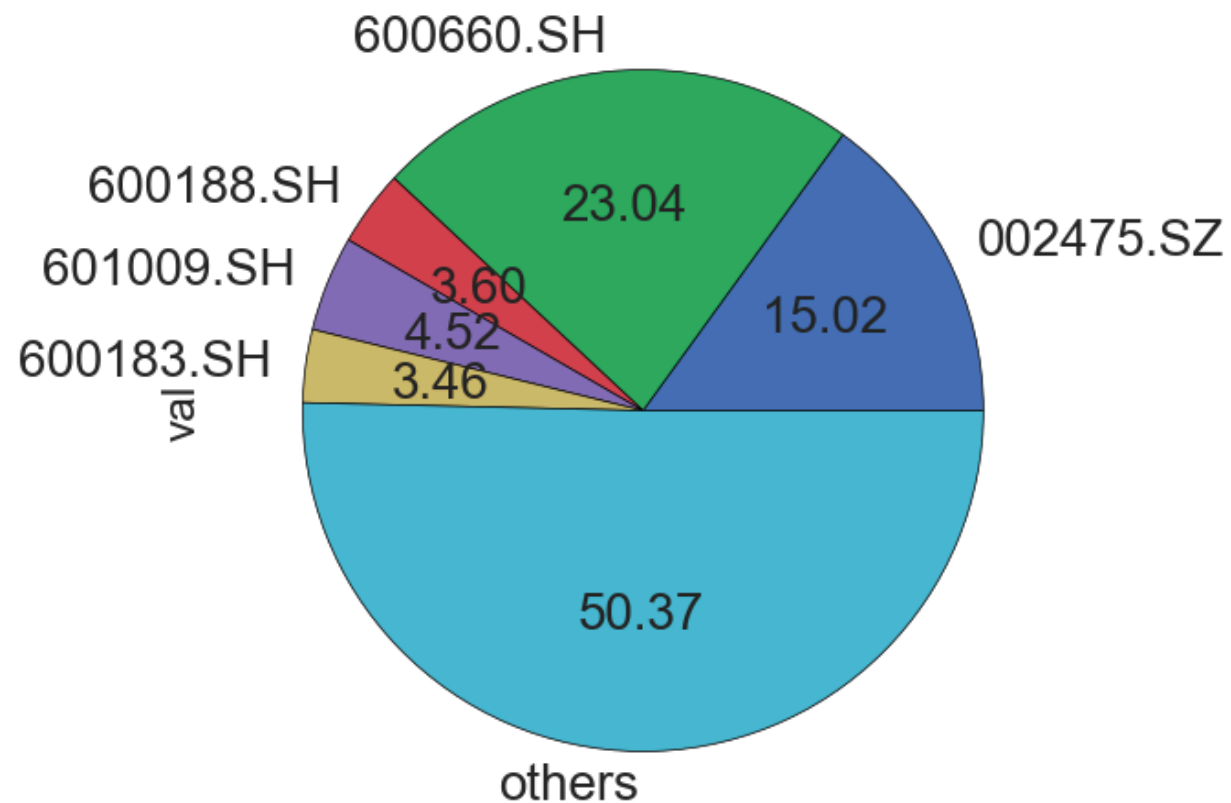
V.US	VISA
USCR.US	US CONCRETE
APPL.US	APPLE
FB.US	FACEBOOK
C.US	CITIGROUP

Heavy-position Assets - HK Stock



01038.HK	長江基建集團
00700.HK	騰訊
02382.HK	舜宇光學科技
02318.HK	中國平安
02689.HK	玖龍紙業

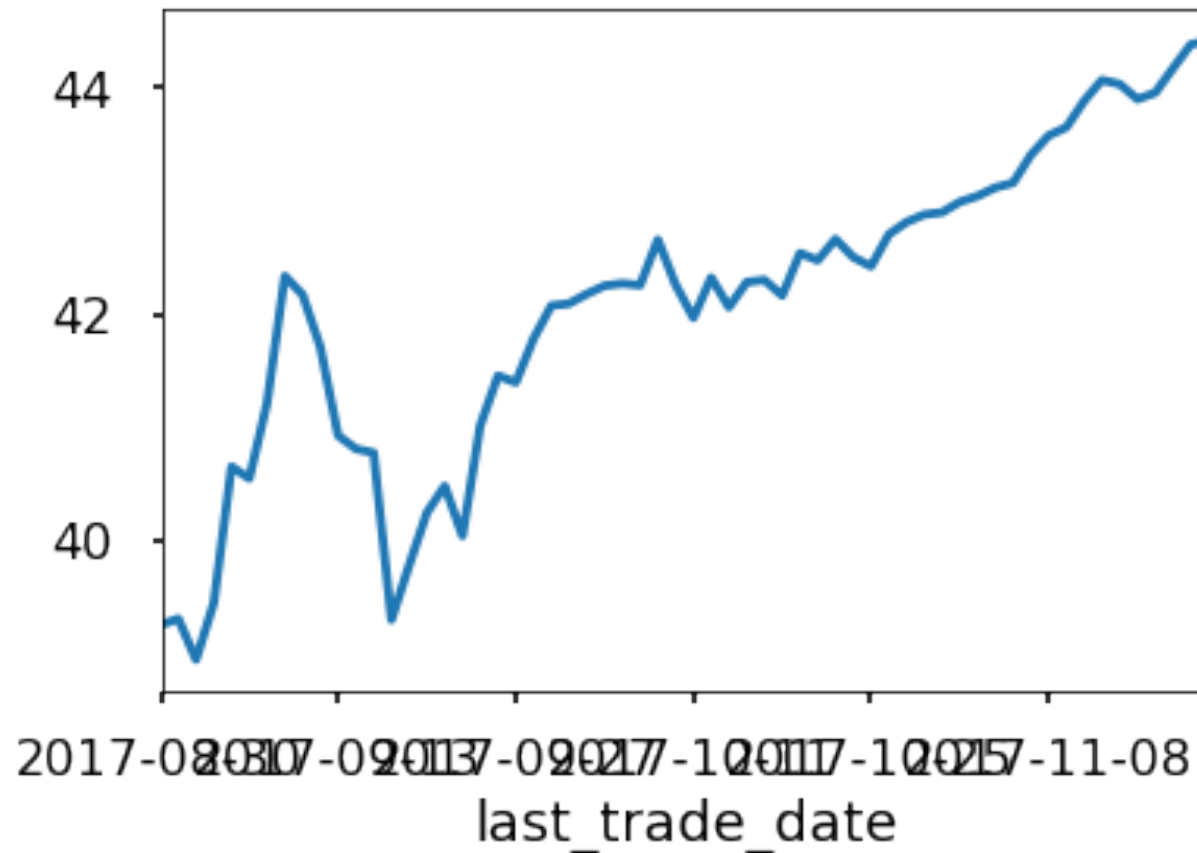
Heavy-position Assets – A share



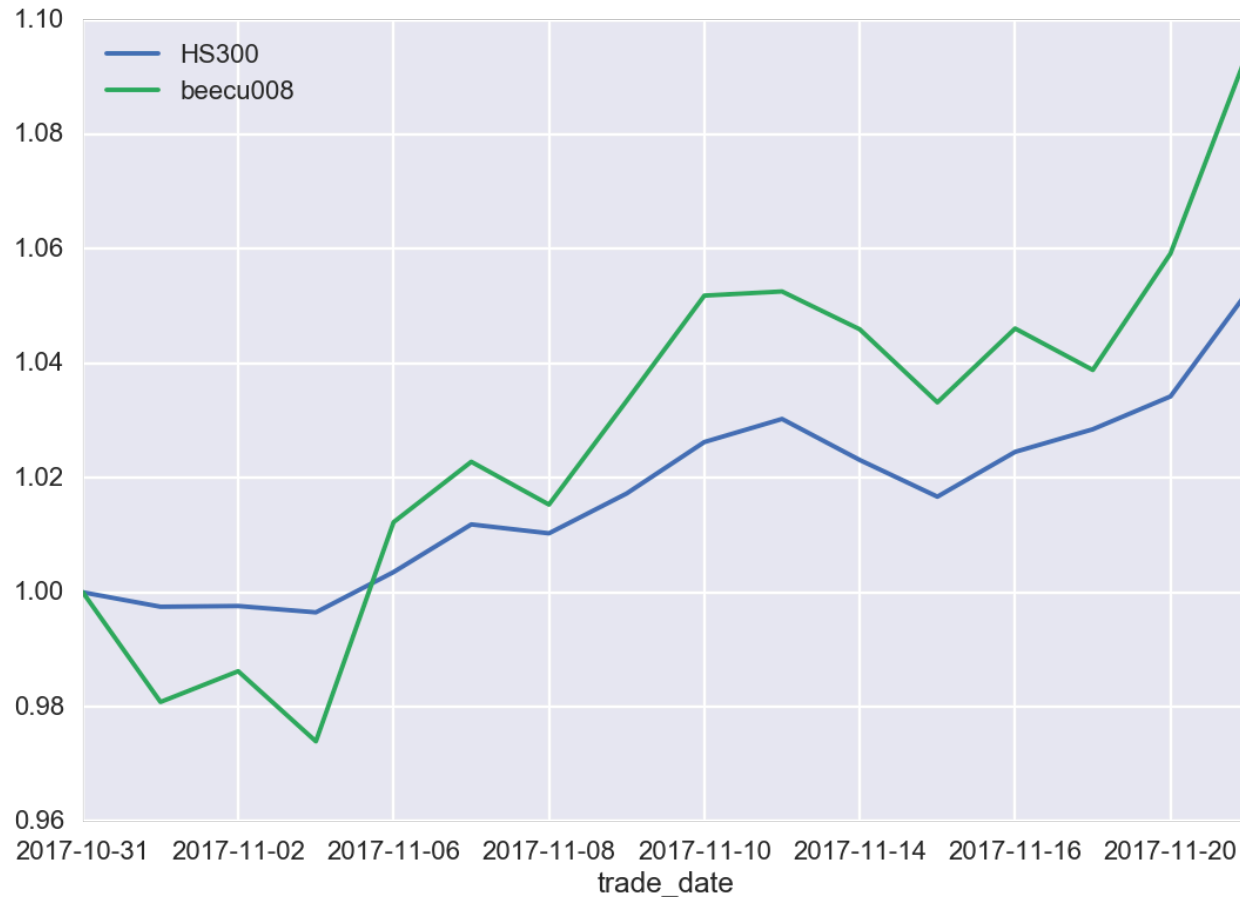
002475.SZ	立讯精密
600660.SH	福耀玻璃
600188.SH	兖州煤业
601009.SH	南京銀行
600183.SH	生益科技

Backup Slides

FDTScore Statistics



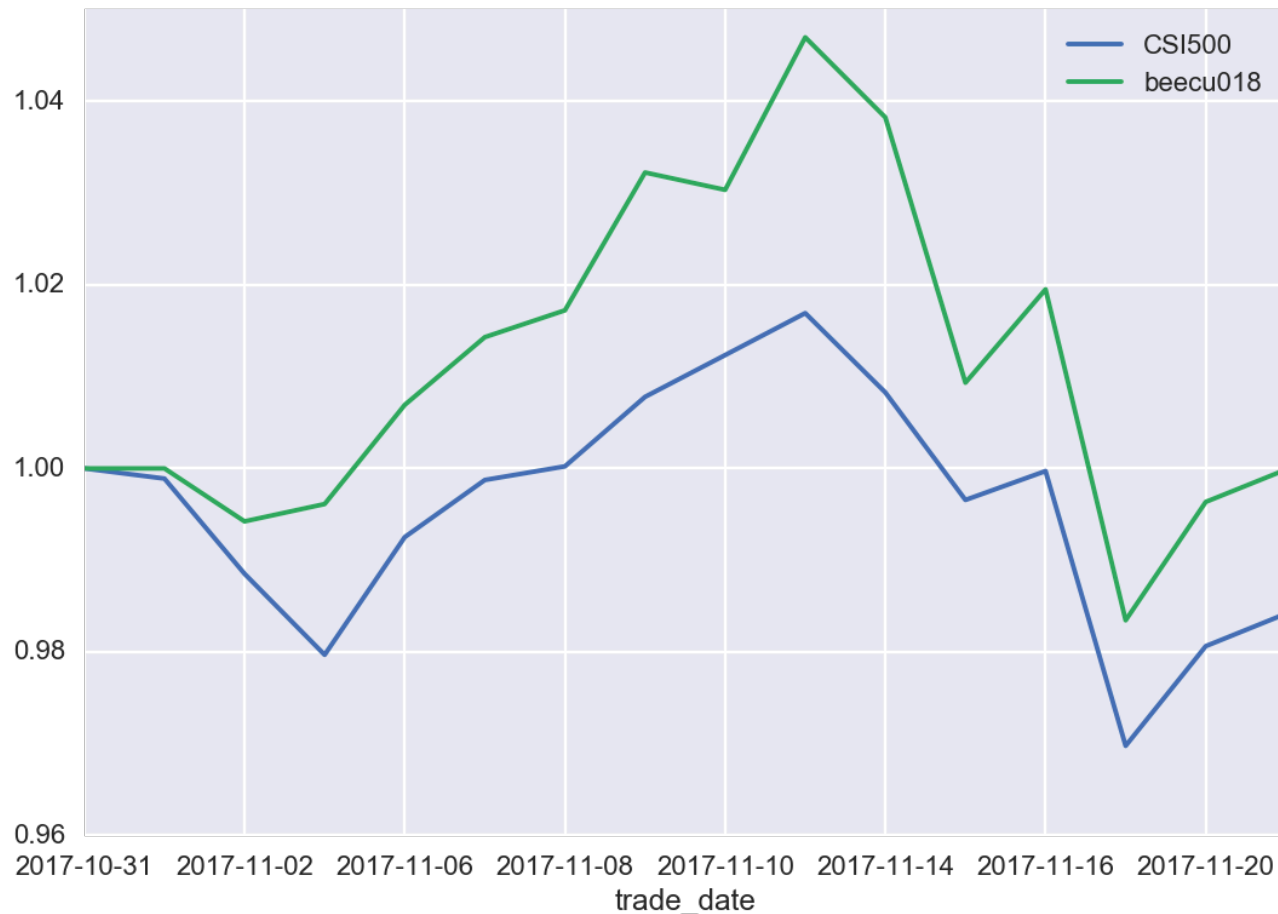
Portfolio Strategy “beecu008”



• Portfolio “beecu008”

- Started on 2017-10-31
- Market: China A
- Benchmark: HS300
- Alpha: -0.09
- Beta: 2.07
- Max Drawdown: -2.6%
- Annual Ret.: 281.8%
- Annual Volatility: 26.0%

Portfolio Strategy “beecu018”



• Portfolio “beecu018”

- Started on 2017-10-31
- Market: China A
- Benchmark: IC500
- Alpha: 0.39
- Beta: 0.98
- Max Drawdown: -6.1%
- Annual Ret.: -0.2%
- Annual Volatility: 22%